Midterm Exam Statistical Mechanics

5 November 2024

Solutions of the exam

Problem Solution: Random energy model (2 points)

Consider a system of N particles. The energy of each particle ϵ_i is positive random variable drawn from an exponential distribution $p(\epsilon) = A\exp(-\beta\epsilon)$, so that the energies of the N particles are independent, identically distributed random variables.

a) What is the expected energy $\langle E_N \rangle$ of a system of N particles?

   First of all we note that $\langle E_N \rangle = \langle \sum_{i=1}^{N} \epsilon_i \rangle$. Then, since the expected value of the sum is the sum of expected values $\langle \sum_{i=1}^{N} \epsilon_i \rangle = \sum_{i=1}^{N} \langle \epsilon_i \rangle$. Since all particles are identical and independent,

   $$\langle \epsilon_i \rangle = \langle \epsilon \rangle = \int_0^\infty d\epsilon \; \epsilon \; p(\epsilon) \qquad \forall i. \tag{1}$$

   Therefore,

   $$\langle E_N \rangle = N \langle \epsilon \rangle \; . \tag{2}$$

   To compute $\langle \epsilon \rangle$, we first compute the normalization constant A by imposing that $\int_0^\infty d\epsilon \; p(\epsilon) = 1$, which yields $A = \beta$. Then we compute

   $$\langle \epsilon_i \rangle = \beta \int_0^\infty d\epsilon \; \epsilon \; e^{-\beta\epsilon} = \frac{1}{\beta} \; . \tag{3}$$

   so that the final answer is $\langle E_N \rangle = N/\beta$.

b) Suppose we consider a system with $N = 2$ particles. What is the distribution of the energy of this system $p(E_2)$? Compute the expected value of E_2 and compare it to the answer to the first question.

   In here, what we want to compute is

   $$p(\epsilon_1 + \epsilon_2 = E_2) = \int d\epsilon_1 \, d\epsilon_2 \, p(\epsilon_1, \epsilon_2) \, \delta\left(E_2 - (\epsilon_1 + \epsilon_2)\right) \tag{4}$$

since the energies are independent random variables $p(\epsilon_1, \epsilon_2) = p(\epsilon_1)p(\epsilon_2)$ so that

$$p(\epsilon_1 + \epsilon_2 = E_2) = \beta^2 \int_0^\infty d\epsilon_1 \int_0^\infty d\epsilon_2 \, e^{-\beta\epsilon_1} e^{-\beta\epsilon_2} \, \delta\left(E_2 - (\epsilon_1 + \epsilon_2)\right) \quad (5)$$

Integrating over ϵ_1 and evaluating the delta function results in

$$p(\epsilon_1 + \epsilon_2 = E_2) = \beta^2 \int_0^{E_2} d\epsilon_2 \, e^{-\beta(E_2 - \epsilon_2)} e^{-\beta\epsilon_2} = \beta^2 \int_0^{E_2} d\epsilon_2 \, e^{-\beta E_2} = \beta^2 \, E_2 e^{-\beta E_2} \, , \quad (6)$$

where note that ϵ_2 is bounded because $\epsilon_1 = E - \epsilon_2 \geq 0$.

Finally, we compute $\langle E_2 \rangle$ as

$$\langle E_2 \rangle = \beta^2 \int_0^\infty dE \, E^2 \, e^{-\beta E} = \frac{2}{\beta} \, , \quad (7)$$

which is exactly the answer to question 1 for $N = 2$.

Problem Solution: Balance of forces (2 points)

Consider a system of n particles, each with mass m, bound by an internal force field, such as gravitational interactions F_i. We define a quantity

$$J = \sum_{i=1}^{n} \vec{p}_i \cdot \vec{r}_i$$

where $\vec{p}_i$ and $\vec{r}_i$ represent the momentum and position of particle i, respectively.

a) Analyze the evolution of J, determining how its rate of change relates to the system's kinetic energy and the interaction between each particle's force and position.

   To analyze the evolution of J, we take its time derivative:

$$\frac{dJ}{dt} = \frac{d}{dt}\left(\sum_{i=1}^{n} \vec{p}_i \cdot \vec{r}_i\right) = \sum_{i=1}^{n}\left(\frac{d\vec{p}_i}{dt} \cdot \vec{r}_i + \vec{p}_i \cdot \frac{d\vec{r}_i}{dt}\right)$$

   Since $\vec{p}_i = m\dot{\vec{r}}_i$, we substitute $\frac{d\vec{p}_i}{dt} = \vec{F}_i$, where $\vec{F}_i$ is the force on the i-th particle. This gives:

$$\frac{dJ}{dt} = \sum_{i=1}^{n}\left(\vec{F}_i \cdot \vec{r}_i + m\dot{\vec{r}}_i \cdot \dot{\vec{r}}_i\right)$$

   The second term, $m\dot{\vec{r}}_i \cdot \dot{\vec{r}}_i$, represents twice the kinetic energy T_i of the i-th particle:

$$m\dot{\vec{r}}_i \cdot \dot{\vec{r}}_i = 2T_i$$

Thus, we can rewrite the time derivative of J as:

$$\frac{dJ}{dt} = \sum_{i=1}^{n} \left(\vec{F}_i \cdot \vec{r}_i + 2T_i \right) = 2T + \sum_{i=1}^{n} \vec{F}_i \cdot \vec{r}_i$$

where $T = \sum_{i=1}^{n} T_i$ is the total kinetic energy of the system. Therefore, the rate of change of J relates to the total kinetic energy and the force-position interaction term.

b) If the system remains stable and confined over a long period, how would the time-averaged rate of change of J behave?

To examine the long-term behavior, we consider the time average of $\frac{dJ}{dt}$ over a long period τ:

$$\left\langle \frac{dJ}{dt} \right\rangle = \frac{1}{\tau} \int_0^{\tau} \frac{dJ}{dt}\, dt$$

If the system remains stable and confined over a long time, the particles' positions and velocities oscillate but remain bounded. Therefore, J does not diverge, and its time derivative oscillates around zero. This means that, in the limit $\tau \to \infty$, the time average of $\frac{dJ}{dt}$ approaches zero:

$$\left\langle \frac{dJ}{dt} \right\rangle = 0$$

c) Based on the previous results, infer the relationship between the system's kinetic energy and the effect of the force field binding the particles. From Part (b), we know that:

$$0 = \left\langle 2T + \sum_{i=1}^{n} \vec{F}_i \cdot \vec{r}_i \right\rangle$$

or equivalently,

$$2\langle T \rangle = -\left\langle \sum_{i=1}^{n} \vec{F}_i \cdot \vec{r}_i \right\rangle$$

This is the *virial theorem*, which states that for a stable, bound system, the time-averaged total kinetic energy $\langle T \rangle$ is related to the time-averaged effect of the force field, represented by $\sum_{i=1}^{n} \vec{F}_i \cdot \vec{r}_i$.

Problem Solution: Ultra-relativistic gas in one dimension in the microcanonical ensemble (3 points)

Consider a non-interacting gas of N ultra-relativistic ideal, indistinguishable particles in a one-dimensional space of volume L. The energy of each particle is $H_i = |p_i|c$, where $|p_i|$ is the absolute value of the momentum of particle i and c is the speed of light.

a) Show that the volume of the M-dimensional convex polyhedron defined by the linear constraint $x_1 + x_2 + x_3 + \cdots + x_M = Q$ with $x_i \geq 0\ \forall i$ is $V_M^Q = Q^M/M!$ [0.5 points].

NOTE: If you do not know how to answer this question, move on and solve the remaining questions.

First, we note that we can express the condition

$$x_1 + x_2 + x_3 + \cdots + x_M = Q \qquad \rightarrow \qquad \frac{x_1}{Q} + \frac{x2}{Q} + \cdots + \frac{x_M}{Q} = 1\ , \tag{8}$$

which means that

$$V_M^Q = \int \prod_i dx_i\, \delta\left(\sum_i x_i - Q\right) = Q^N \int \prod_i d\tilde{x}_i\, \delta\left(\sum_i \tilde{x}_i - 1\right) = Q^N V_M^1 . \tag{9}$$

Then we realize that for $M = 1$, $V_1^r = r$, for $M = 2$,

$$V_2^1 = \int_0^1 dx_2 V_1^{x_2} = \frac{1}{2}\ . \tag{10}$$

Similarly for $M = 3$:

$$V_3^1 = \int_0^1 dx_3 V_2^{x_3} = \frac{1}{3!}\ . \tag{11}$$

In the end, in general we get that

$$V_m^1 = \int_0^1 dx_m V_{m-1}^{x_2} = \frac{1}{m!}\ . \tag{12}$$

So that $V_M^Q = Q^M/M!$.

b) Find the number of microstates compatible with a total energy E and its associated entropy for large N as a function of N, L, and c.

Since the energy does not depend on the position coordinates, we can compute the number of microstates as,

$$\Omega(E, L, N) = \Omega(\mathrm{Q})\Omega(\mathrm{P}) = \frac{1}{N!}\left(\frac{L}{h}\right)^N \int \prod_i dp_{x,i}\, \delta\left(\sum_i |p_{x,i}|c - E\right) \tag{13}$$

where note that we have introduced the Gibbs factor $N!$ because particles are indistinguishable and $\Omega(Q) = \int \frac{dQ}{h^N} = \left(\frac{L}{h}\right)^N$ because all particles can occupy the same volume L.

To compute the integral, first we note that the integral is symmetric with respect to zero for each $p_{x,i}$, therefore we can assume that $\int_{-\infty}^{\infty} dp_x = 2\int_0^{\infty} dp_x$, so that the constrain is strictly over positive variables:

$$\begin{aligned}\int \prod_i dp_{x,i}\, \delta\left(\sum_i |p_{x,i}|c - E\right) &= 2^N \prod_i \int_0^{\infty} dp_{x,i}\, \delta\left(\sum_i p_{x,i}c - E\right) \\ &= \left(\frac{2}{c}\right)^N \int_0^{\infty} d\tilde{p}_{x,i}\, \delta\left(\sum_i \tilde{p}_{x,i} - E\right) ,\end{aligned}$$

where we have defined $\tilde{p} = p\,c$. The integral then corresponds to the surface of the polyhedron defined by the condition $\sum_i \tilde{p}_{x,i} = E$, which has a volume $V_N^E = E^N/N!$ The surface of this polyhedron is the derivative of this volume with respect to the energy so that

$$\int \prod_i dp_{x,i}\, \delta\left(\sum_i |p_{x,i}|c - E\right) = \left(\frac{2}{c}\right)^N \frac{d}{dE} V_N^E = \left(\frac{2E}{c}\right)^N \frac{N}{E\,N!},$$

. So that the total number of configurations is

$$\Omega(E, L, N) = \left(\frac{2\,E\,L}{h\,c}\right)^N \frac{N}{E\,N!\,N!} , \tag{14}$$

and the entropy in the large N limit is:

$$S = k_B \log \Omega(E, L, N) = k_B N \log\left(\frac{E\,L}{h\,c\,N^2}\right) + 2\,k_B\,N\ . \tag{15}$$

c) Find how the energy of the system depends on temperature and compute the specific heat of the system. Does it satisfy the equipartition theorem? Justify your answer.

To find how the enrgy depends on temperature we use the relationsjip $\frac{\partial S}{\partial E} = \frac{1}{T}$ to obtain

$$\frac{1}{T} = \frac{N k_B}{E} \qquad \rightarrow \qquad E = N\,k_B\,T\ , \tag{16}$$

which does not satisfy the equipartition theorem. The reason for this is that de dependency of the Hamiltonian on the momenta is linear and not quadratic.

d) Find the equation of state and compare it to the equation of state of an ideal gas. Does your result contradict your answer to the second question?

To find the equation of state, we use the relationship $\frac{\partial S}{\partial L} = \frac{P}{T}$ and find

$$\frac{P}{T} = \frac{N\,k_B}{L} \qquad \rightarrow \qquad P\,L = N\,k_B\,T\ , \tag{17}$$

which is the equation of state of an ideal, non-interacting gas. This result does not contradict the result in the previous question because it stems from the fact that the particles are non-interacting and is independent of the specific shape of the Hamiltonian.

Problem Solution: Ideal gas in a surface (3 points)

An ideal gas of N classical point particles is confined to the two-dimensional surface of a sphere. Determine the internal energy $U(T, A)$, where A is the surface area, and derive the equation of state for the system on the spherical surface.
The position $\vec{r}$ of a point on the surface of a sphere with radius r is given by:

$$\vec{r} = r\sin\theta\cos\phi\,\hat{i} + r\sin\theta\sin\phi\,\hat{j} + r\cos\theta\,\hat{k}$$

where θ is the polar angle and ϕ is the azimuthal angle.
Hint: Recall classical mechanics. The Lagrangian L of a single particle in an ideal gas is $L = \frac{1}{2}m|\dot{\mathbf{r}}|^2$. For each generalized coordinate q_i, the conjugate momentum p_i is defined as

$$p_i = \frac{\partial L}{\partial \dot{q}_i}$$

The Hamiltonian H in generalized coordinates is given by

$$H = \sum_i p_i\dot{q}_i - L$$

To solve the problem, we first compute the partition function in the surface of a sphere. The Lagrangian L for a particle of mass m moving on a two-dimensional surface (either a sphere or torus) is kinetic:

$$L = \frac{1}{2}m|\dot{\vec{r}}|^2$$

For a particle on the surface of a sphere of radius r, we use spherical coordinates (θ, ϕ), where θ is the polar angle and ϕ is the azimuthal angle. The position vector $\vec{r}$ is:

$$\vec{r} = r\sin\theta\cos\phi\,\hat{i} + r\sin\theta\sin\phi\,\hat{j} + r\cos\theta\,\hat{k}$$

To compute the velocity vector $\dot{\vec{r}}$, we take the time derivative of each component of $\vec{r}$. Since r is constant, we differentiate with respect to θ and ϕ as they vary with time.

$$\dot{\vec{r}} = \frac{d}{dt}\left(r\sin\theta\cos\phi\,\hat{i} + r\sin\theta\sin\phi\,\hat{j} + r\cos\theta\,\hat{k}\right)$$

Expanding each term:

$$\frac{d}{dt}(r\sin\theta\cos\phi) = r\cos\theta\,\dot{\theta}\cos\phi - r\sin\theta\sin\phi\,\dot{\phi}$$

$$\frac{d}{dt}(r\sin\theta\sin\phi) = r\cos\theta\,\dot{\theta}\sin\phi + r\sin\theta\cos\phi\,\dot{\phi}$$

$$\frac{d}{dt}(r\cos\theta) = -r\sin\theta\,\dot{\theta}$$

To find $|\dot{\vec{r}}|^2$, we take the dot product of $\dot{\vec{r}}$ with itself:

$$|\dot{\vec{r}}|^2 = \dot{\vec{r}} \cdot \dot{\vec{r}}$$

Expanding this dot product:

$$|\dot{\vec{r}}|^2 = \left(r\cos\theta\,\dot{\theta}\cos\phi - r\sin\theta\sin\phi\,\dot{\phi}\right)^2 + \left(r\cos\theta\,\dot{\theta}\sin\phi + r\sin\theta\cos\phi\,\dot{\phi}\right)^2 + \left(-r\sin\theta\,\dot{\theta}\right)^2$$

Expanding each term:

$$\left(r\cos\theta\,\dot{\theta}\cos\phi - r\sin\theta\sin\phi\,\dot{\phi}\right)^2 = r^2\cos^2\theta\,\dot{\theta}^2\cos^2\phi + r^2\sin^2\theta\,\dot{\phi}^2\sin^2\phi - 2r^2\cos\theta\sin\theta\,\dot{\theta}\dot{\phi}\cos\phi\sin\phi$$

$$\left(r\cos\theta\,\dot{\theta}\sin\phi + r\sin\theta\cos\phi\,\dot{\phi}\right)^2 = r^2\cos^2\theta\,\dot{\theta}^2\sin^2\phi + r^2\sin^2\theta\,\dot{\phi}^2\cos^2\phi + 2r^2\cos\theta\sin\theta\,\dot{\theta}\dot{\phi}\cos\phi\sin\phi$$

$$\left(-r\sin\theta\,\dot{\theta}\right)^2 = r^2\sin^2\theta\,\dot{\theta}^2$$

Adding these terms together, we see that the cross terms cancel, leaving:

$$|\dot{\vec{r}}|^2 = r^2\cos^2\theta\,\dot{\theta}^2(\cos^2\phi + \sin^2\phi) + r^2\sin^2\theta\,\dot{\phi}^2(\sin^2\phi + \cos^2\phi) + r^2\sin^2\theta\,\dot{\theta}^2$$

Since $\cos^2\phi + \sin^2\phi = 1$, we obtain:

$$|\dot{\vec{r}}|^2 = r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2$$

This is the expression for the velocity squared on the surface of a sphere.
Thus, the Lagrangian L becomes:

$$L = \frac{1}{2}mr^2\left(\dot{\theta}^2 + \sin^2\theta\,\dot{\phi}^2\right)$$

And then, the conjugate momenta p_θ and p_ϕ are:

$$p_\theta = \frac{\partial L}{\partial\dot{\theta}} = mr^2\dot{\theta}$$

$$p_\phi = \frac{\partial L}{\partial\dot{\phi}} = mr^2\sin^2\theta\,\dot{\phi}$$

The Hamiltonian for a single particle H_1 is:

$$H_1 = p_\theta\dot{\theta} + p_\phi\dot{\phi} - L$$

Substituting $\dot{\theta} = \frac{p_\theta}{mr^2}$ and $\dot{\phi} = \frac{p_\phi}{mr^2\sin^2\theta}$:

$$H_1 = \frac{p_\theta^2}{2mr^2} + \frac{p_\phi^2}{2mr^2 \sin^2\theta}$$

For an ideal gas of N non-interacting particles, the canonical partition function Z is:

$$Z = \frac{1}{N!}\left(Z_1\right)^N$$

where Z_1 is the single-particle partition function.
The single-particle partition function Z_1 is the product of the position and momentum integrals:

$$Z_1 = \int d^2q \int d^2p\, e^{-\beta H_1}$$

where $\beta = \frac{1}{k_B T}$ and H_1 is the Hamiltonian derived above.
Let us write the partition function:

$$\begin{aligned}
Z_1 &= \int \exp\left[-(\frac{1}{2mr^2kT}p_\theta^2 + \frac{1}{2mr^2kT\sin^2\theta}p_\phi^2)\right] dp_\theta dp_\phi d\theta d\phi && (18)\\
&= \int (2\pi mr^2kT)^{\frac{1}{2}} (2\pi mr^2kT\sin^2\theta)^{\frac{1}{2}}\, d\theta d\phi && (19)\\
&= (2\pi mr^2kT)2\pi \int_0^\pi |\sin\theta| d\theta && (20)\\
&= 2\pi mAkT && (21)\\
& && (22)
\end{aligned}$$

where $A = 4\pi r^2$ is the area of the sphere.
The total partition function is then:

$$Z = \frac{1}{N!}\left(Z_1\right)^N = \frac{1}{N!}(2\pi mAkT)^N \qquad (23)$$

The internal energy U in the canonical ensemble is given by:

$$U = kT^2 \frac{\partial}{\partial T}\ln Z$$

$$\ln Z = -\ln N! + N\left(\ln(2\pi mAk) + N\ln T\right)$$

Differentiating with respect to T:

$$U = kT^2\frac{\partial}{\partial T}\ln Z = kT^2\frac{N}{T} = NkT$$

The pressure P in the canonical ensemble is given by:

$$P = \frac{\partial}{\partial A}(kT\ln Z)$$

note that here the role of the volume is played by the area! Then:

$$P = (kT\frac{\partial}{\partial A}\ln Z) = \frac{NkT}{A}$$